Hankuk University
Department of Mathematics
MATH 101 Calculus I
Credits: 3 (3 hours lecture per week)

Course Description
Differential and integral calculus of functions of one variable for science and engineering majors: limits, continuity, derivatives, applications of the derivative, and the definite integral.

Learning Outcomes
1. Evaluate limits algebraically and graphically, including one-sided limits and limits at infinity.
2. Compute derivatives using the product rule, quotient rule and chain rule, and apply implicit differentiation.
3. Solve related rates and optimization problems.
4. Evaluate definite integrals using the Fundamental Theorem of Calculus and substitution.

Textbook
James Stewart, Calculus: Early Transcendentals, 9th edition

Prerequisites
High school calculus or placement exam

Weekly Schedule
Week 1: Functions and models, review of trigonometric functions
Week 2: The limit of a function, one-sided limits
Week 3: Continuity, intermediate value theorem, limits at infinity
Week 4: Derivatives and rates of change, the derivative as a function
Week 5: Differentiation rules: product rule and quotient rule
Week 6: The chain rule, implicit differentiation
Week 7: Related rates, linear approximation
Week 8: Midterm exam
Week 9: Maximum and minimum values, the mean value theorem
Week 10: Curve sketching, L'Hopital's rule
Week 11: Optimization problems
Week 12: Antiderivatives, Riemann sums and the definite integral
Week 13: The fundamental theorem of calculus
Week 14: The substitution rule
Week 15: Final exam

Assessment
Homework 20%, Quizzes 10%, Midterm 30%, Final exam 40%